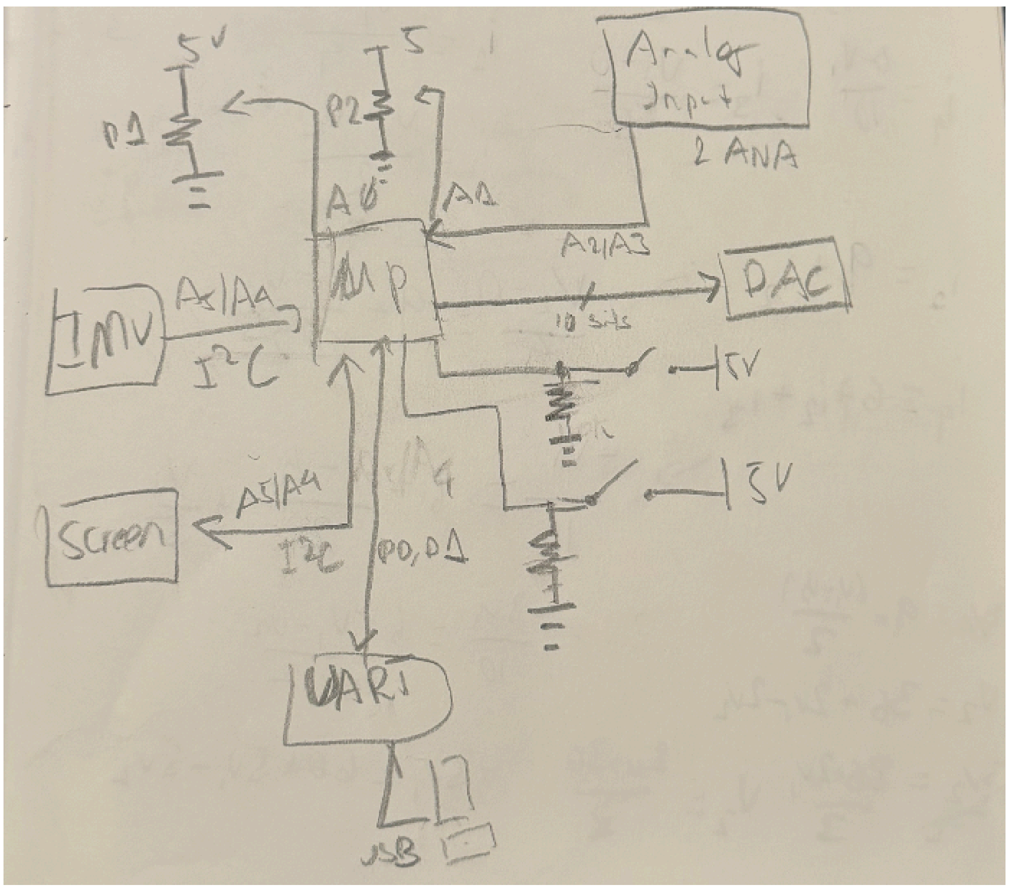
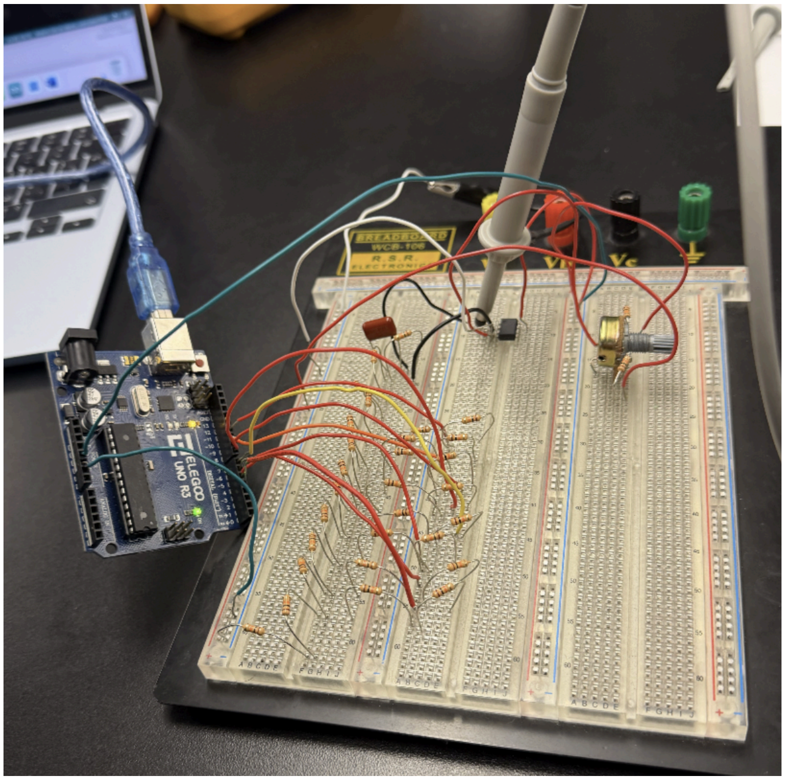
Introduction
In this project, we were assigned the task of designing and building and Arduino Uno shield. We especially made a DSP device with a 10bit DAC, and used KiCAD for the PCB design. The purpose of this project was familiarization with the tools, methodologies, and process of building and designing microprocessors.

Design Stage

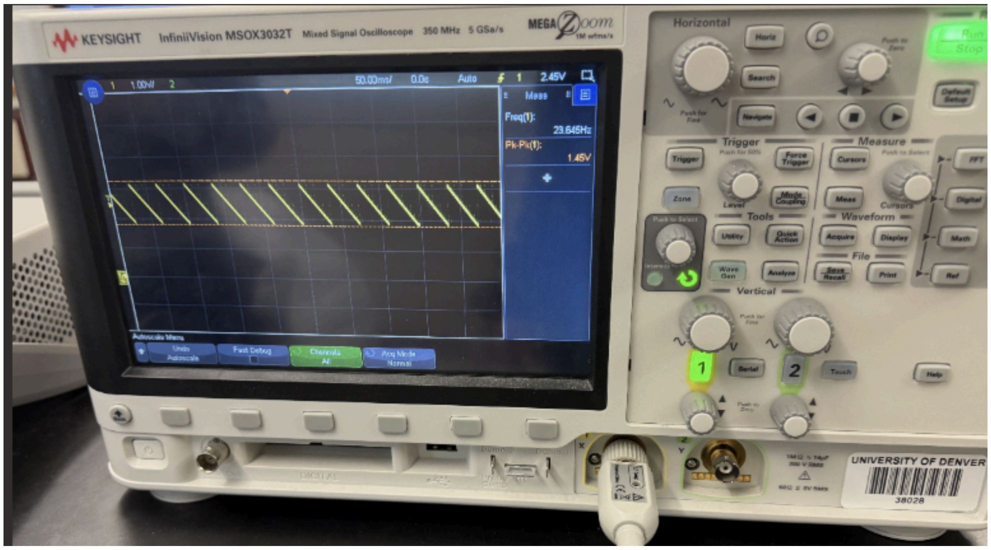
To start the process, we designed a block diagram for the shield, consisting of a 10bit DAC, OLED screen, IMU, Analog Input, UART, three potentiometers, two analog inputs, and two buttons alongside the main microcontroller.



Prototype Stage
Before fully designing the shield, we first constructed a R2R ladder DAC. This was done on a breadboard, and we were also assigned with coding so that we may see output on an oscilloscope.

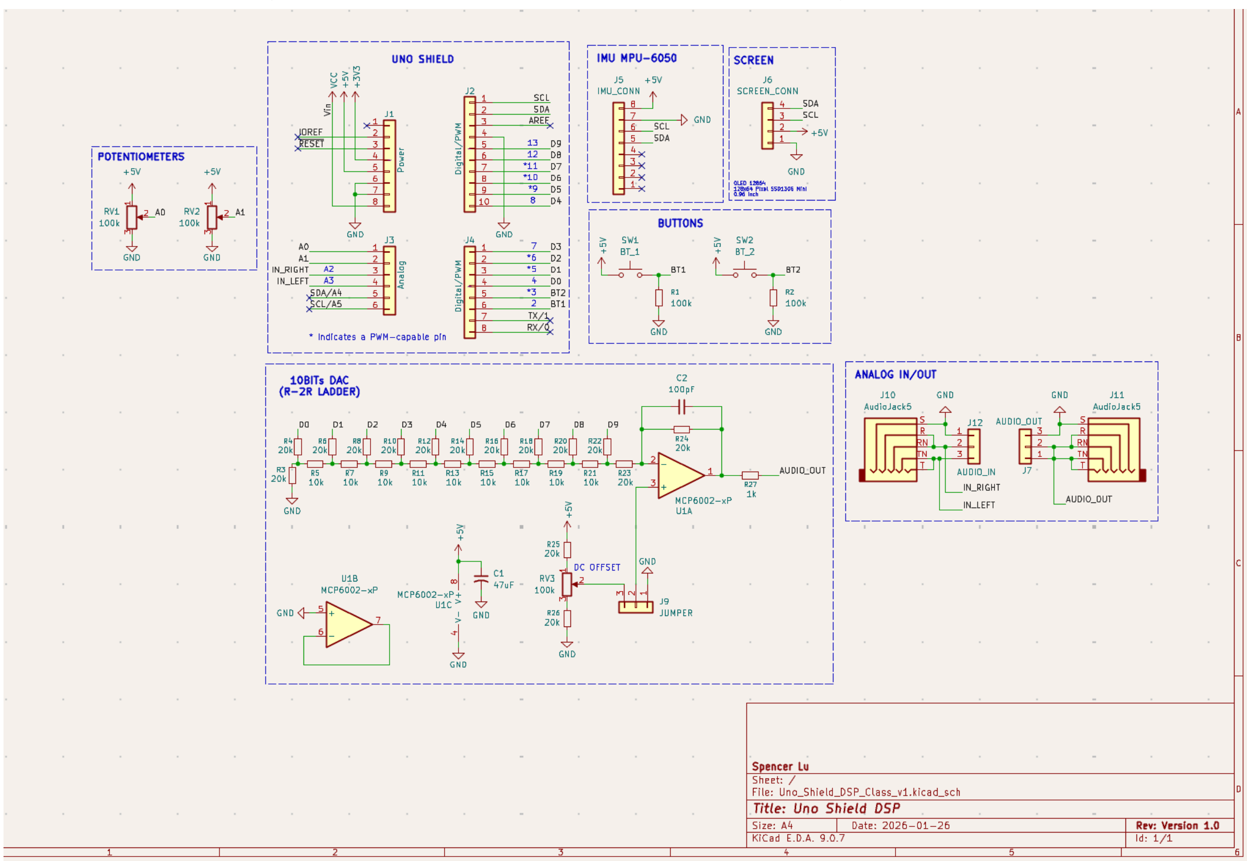

BREADBOARD
WCB-106
R.S.R.
ELEGOO
UNO R3

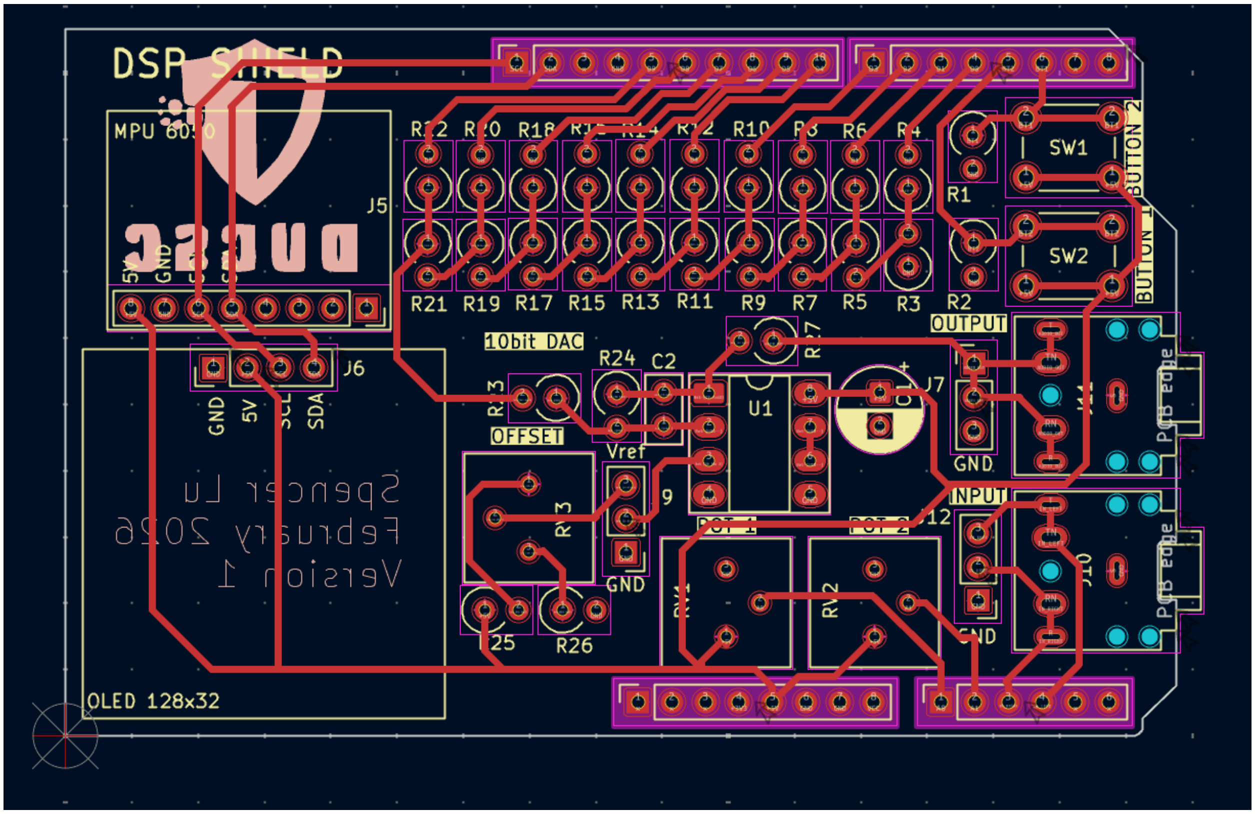
```
/*
Scope test for Phase B prototype:
- 10-bit R-2R DAC data bus: D4..D13
- Outputs a repeating ramp 0..1023 to the R-2R ladder
- Also outputs a 1 kHz square wave on D3 for quick scope sanity check

How to use:
1) Probe square wave: scope tip on D3, ground on Arduino GND -> should see 1 kHz square wave.
2) Probe DAC output: scope tip on the analog output of your R-2R ladder (or op-amp output),
ground on Arduino GND -> should see a ramp/staircase waveform.
*/

#include <Arduino.h>

const uint8_t SQUARE_PIN = 3; // quick test output
const uint8_t dacPins[10] = {4,5,6,7,8,9,10,11,12,13}; // 10-bit bus

// Write 10-bit value (0..1023) to R-2R ladder
void writeDAC10(uint16_t value)
{
value &= 0x03FF; // keep only 10 bits

// D4 = bit0 (LSB), D13 = bit9 (MSB)
for (int bit = 0; bit < 10; bit++)
{
digitalWrite(dacPins[bit], (value >> bit) & 1);
}
}

void setup()
{
pinMode(SQUARE_PIN, OUTPUT);
digitalWrite(SQUARE_PIN, LOW);

for (int i = 0; i < 10; i++)
{
pinMode(dacPins[i], OUTPUT);
digitalWrite(dacPins[i], LOW);
}
}

void loop()
{
// ----- 1 kHz square wave on D3 (sanity check) -----
// 1 kHz period = 1000 us, half period = 500 us
digitalWrite(SQUARE_PIN, HIGH);
delayMicroseconds(500);
digitalWrite(SQUARE_PIN, LOW);
delayMicroseconds(500);

// ----- DAC ramp 0..1023 -----
// One ramp step per loop call would be too slow because of the square wave delays,
// so we do a short burst of DAC steps each time.
for (uint16_t code = 0; code < 1024; code++)
{
writeDAC10(code);

// Controls ramp speed:
// - smaller = faster (higher output frequency)
// - bigger = slower (easier to see stair-steps)
delayMicroseconds(80);
}
}
```

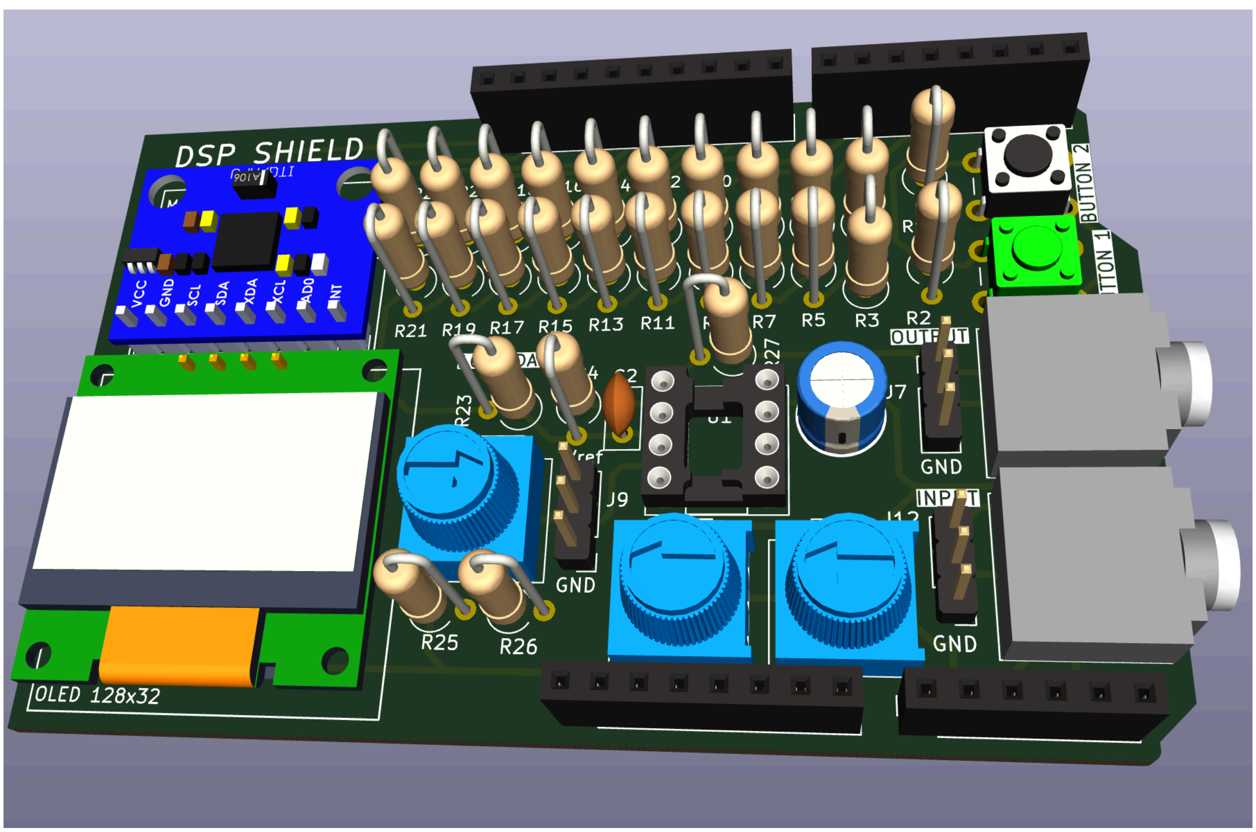


PCB Design

After the initial prototype, we then constructed the schematics, layout, and 3D render in KiCad.




DSP SHIELD
MPU 6050
J5
5V
GND
J6
GND
5V
SCL
SDA
R22 R20 R18 R16 R14 R12 R10 R8 R6 R4
R21 R19 R17 R15 R13 R11 R9 R7 R5 R3
R1
R2
SW1
SW2
BUTTON 2
OUTPUT
10bit DAC
R24
C2
R27
R23
OFFSET
Vref
U1
J7
GND
J9
GND
POT 1
POT 2
J12
INPUT
GND
RV3
RV1
RV2
R25
R26
J10
PCB edge
Spencer Lu
February 2026
Version 1
OLED 128x32


DSP SHIELD
VCC
GND
SCL
SDA
XDA
XCL
AD0
INT
R21 R19 R17 R15 R13 R11
R7 R5 R3 R2
OUTPUT
C2
U1
J7
GND
J9
GND
INPUT
GND
R25
R26
BUTTON 2
OLED 128x32

Assemble Stage

We then assembled the boards by soldering the pieces individually.